

Drill Problems 9

*Author: Jaehoon Song**Release: 2025-08-10 15:08:31-04:00***Purpose and Usage:**

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**C O L U M B I A A C A D E M Y***enrichment beyond the classroom*

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Written by Jaehoon Song

1. **System of Equations Solutions** (10 points)

How many solutions does the given system of equations have?

$$\begin{aligned}y &= 2x - 5 \\ y &= 2x^2 - 18x + 45\end{aligned}$$

- A. 0
- B. 1
- C. 2
- D. ∞

Answer:



2. **Ice Cream Sundae Toppings** (10 points)

The function $c(t) = 6.50 + 1.50t$ can be used to find the cost of an ice cream sundae, where t is the number of toppings added to the sundae. If Henry ordered an ice cream sundae, paid for it with \$20, and received \$6 in change, how many toppings did Henry put on his sundae?

- A. 3
- B. 5
- C. 7
- D. 9

Answer:



3. **Sugar in Soda** (10 points)

There are 3 teaspoons in a tablespoon. There are 4 tablespoons in $\frac{1}{4}$ cup. If there are 10 teaspoons of sugar in one can of soda, how many cups of sugar are in 12 cans of soda?

- A. $\frac{5}{6}$
- B. $2\frac{1}{2}$
- C. 4
- D. 10

Answer:



4. **Trigonometric Identity** (10 points)

Angle R in a right triangle measures n° . If $\sin n^\circ = \frac{5}{13}$, what is $\cos(90 - n)^\circ$?

Answer:



5. **Perpendicular Line y -Intercept** (10 points)

Line l is perpendicular to the line $-4x + 2y = 14$ and passes through the point $(4, 6)$. What is the coordinate of the y -intercept of line l ?

- A. $(0, 4)$
- B. $(0, 5)$
- C. $(0, 7)$
- D. $(0, 8)$

Answer:



6. **Bouquet of Flowers** (10 points)

Sandra is putting together a bouquet of flowers using tulips and roses. She can have a total of 24 flowers in the bouquet. Each tulip costs \$4 and each rose costs \$6. If she spends \$126 on the bouquet, how many roses does Sandra put in the bouquet?

- A. 9
- B. 12
- C. 15
- D. 18

Answer:



7. **Kinetic Energy Formula** (10 points)

The formula for finding the kinetic energy of an object is $K = \frac{1}{2}mv^2$, where K is the kinetic energy of the object in joules, m is the mass of the object in kilograms, and v is the velocity of the object in meters per second. Which equation correctly expresses v in terms of K and m ?

- A. $v = \pm\sqrt{\frac{2K}{m}}$
- B. $v = \pm\sqrt{\frac{K}{2m}}$
- C. $v = \pm\sqrt{\frac{2m}{K}}$
- D. $v = \pm\sqrt{\frac{1}{2}Km}$

Answer:



8. **Quadratic Vertex Value** (10 points)

For a quadratic function $f(x) = x^2 + bx + c$, the value $f(3) = -4$ is the vertex of the function. What is the value of c ?

Answer:

□

9. **Pizza Price Scatterplot** (10 points)

The scatterplot shows the relationship between the diameter of pizza in inches at a restaurant and the price of the pizza per square inch in dollars.

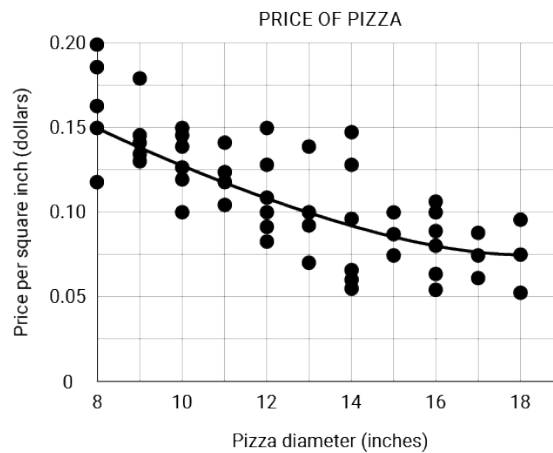


Figure 1: reference attached

Which of the following is the best estimate of the diameter, in inches, of a pizza that costs \$0.12 per square inch?

Answer:

□

10. **Circle Center** (10 points)

The equation of the circle in the xy -plane is given:

$$x^2 + y^2 - 6x + 12y = 3$$

What is the center of the circle?

- A. $(-6, 12)$
- B. $(-3, 6)$
- C. $(3, -6)$
- D. $(6, -12)$

Answer:

□

11. **Solving for x** (10 points)

If $4x + 16 = 24$, what is the value of $x + 4$?

- A. 2
- B. 4
- C. 6
- D. 8

Answer:



12. **Percent of a Value** (10 points)

If x is $2k\%$ of y , what is the value of $3k\%$ of $0.3y$?

- A. $0.05x$
- B. $0.2x$
- C. $0.45x$
- D. $0.9x$

Answer:



13. **Cubic Function Evaluation** (10 points)

For the given function $h(x) = 3x^3 - ax$, $h(2) = 20$. What is the value of $h(-1)$?

- A. -5
- B. -1
- C. 1
- D. 5

Answer:



14. **Trigonometric Relationship** (10 points)

An angle opposite a leg of a right triangle measures w° , and $\tan(w^\circ) = \frac{3}{4}$. What is $\sin(90^\circ - w^\circ)$?

Answer:



15. **Exponent Rules** (10 points)

Which of the following is equivalent to $x^{0.6} \cdot x^{\frac{3}{20}}$ for all positive values of x ?

- A. $\sqrt[20]{x^{0.9}}$
- B. $\sqrt[5]{x^3}$
- C. $\sqrt[4]{x^3}$
- D. $\sqrt[3]{x^4}$

Answer:



16. **System of Equations: Value of $a - b$** (10 points)

$$3.5a - 4.25b = -14.25$$

$$4.25a - 3.5b = -9$$

The solution to the given system of equations is (a, b) . What is the value of $a - b$?

- A. -7
- B. -3
- C. 5
- D. 7

Answer:



17. **Mean vs. Median** (10 points)

A survey counted the number of flowers on the bushes in a park. The mean number of flowers per bush was 34 and the median number was 45. What factor most likely would account for the difference between the mean and the median?

- A. There were some bushes with extremely few flowers.
- B. The mode of the data was a number greater than 34.
- C. Many bushes had between 34 and 45 flowers.
- D. There is little variance in the number of flowers on the bushes.

Answer:



18. **Coffee Preferences Table** (10 points)

The table shows the distribution of coffee preferences among the 27 members of the DC Book Club. If a

	Cream	No Cream	Total
Sugar	4	7	11
No Sugar	15	1	16
Total	19	8	27

club member is selected at random, what is the probability the member prefers no cream, given that they prefer sugar?

- A. $\frac{7}{27}$
- B. $\frac{7}{16}$
- C. $\frac{7}{11}$
- D. $\frac{7}{8}$

Answer:



19. **Point on a Line** (10 points)

A line in the xy -plane contains the point $(3, -5)$ and has a slope of $\frac{4}{3}$. Which of the following points lies on the line?

- A. $(\frac{1}{2}, 1\frac{2}{3})$
- B. $(1\frac{2}{3}, -5)$
- C. $(2, -6\frac{1}{3})$
- D. $(5, -5\frac{1}{4})$

Answer:



20. **Volume of a Cone** (10 points)

The height of a cone is three times its diameter. The base of the cone is a circle, and the area of the base is 9π square centimeters. What is the volume of the cone, in cubic centimeters?

- A. 27π
- B. 54π
- C. 162π
- D. 216π

Answer:



21. **Compound Interest Formula** (10 points)

A formula used for compound interest is $A = P \left(1 + \frac{r}{n}\right)^{nt}$. Which of the following represents P in terms of A , n , and t ?

- A. $\frac{(1 + \frac{r}{n})^{nt}}{A}$
- B. $A \left(1 + \frac{r}{n}\right)^{nt}$
- C. $\frac{A}{(1 + \frac{r}{n})^{nt}}$
- D. $\frac{1}{A(1 + \frac{r}{n})^{nt}}$

Answer:



22. **Parking Garage Usage** (10 points)

A parking garage charges \$11 on weekdays and \$15 on weekends. If the garage brought in \$53,234 in one week and a total of 4,122 people used the garage, how many people used the garage during the weekend?

- A. 304
- B. 1,973
- C. 2,149
- D. 3,549

Answer:



23. **Exponential Function Value** (10 points)

The exponential function f is defined by $f(x) = 11 \cdot b^x$ where b is a positive constant. If $f(4) = 2,816$, what is the value of $f(5)$?

- A. 4
- B. 11,264
- C. 161,051
- D. 164,916,224

Answer:



24. **Solving for a and b** (10 points)

$$a(x + 4) = b(x - 1) + 1$$

If the solution to the given equation is $x = 4$, which of the following could be a value of a and b ?

- A. $a = 8$ and $b = -3$
- B. $a = 7$ and $b = 6$
- C. $a = 4$ and $b = -5$
- D. $a = -4$ and $b = -11$

Answer:



25. **Exponential Equation** (10 points)

If $\frac{a^{2x^3}}{a^4} = a^{12}$, what is the value of 2^x ?

Answer:



26. **Quadratic Expansion and Coefficient** (10 points)

If $(3x + a)(2x + b) = 6x^2 + cx + 6$ and $4a + 6b = 36$ for all values of x , what is the value of c ?

Answer:



27. **Intersection of Lines** (10 points)

The graph of line q is perpendicular to the line $2x - 4y = 11$ and contains the point $(1, 6)$. Line p contains the points $\left(2\frac{1}{4}, 7\frac{1}{4}\right)$ and $\left(\frac{1}{3}, 5\frac{1}{3}\right)$. If p and q intersect at the point (a, b) , what is the value of $a - b$?

- A. -5
- B. 1
- C. 5
- D. 6

Answer:



28. **Triangle Congruence** (10 points)

In triangles ABC and DEF , $\angle A$ and $\angle D$ each measure 48° , and the measures of $\overline{AB}$ and $\overline{DE}$ are both 5 inches. Which of the following is sufficient to prove that ABC and DEF are congruent triangles?

- A. $\overline{BC}$ measures 3 inches and $\overline{EF}$ measures 4 inches.
- B. $\overline{BC}$ measures 4 inches and $\overline{EF}$ measures 4 inches.
- C. $\overline{AC}$ measures 4 inches and $\overline{DF}$ measures 4 inches.
- D. $\overline{BC}$ measures 3 inches and $\overline{DF}$ measures 4 inches.

Answer:

□

29. **Quadratic Function Maximum** (10 points)

$$f(x) = -2x^2 + 16x - 11$$

If $g(x) = f(x - 3)$, what is the value of x where $g(x)$ reaches its maximum value?

Answer:

□

30. **Algebraic Expression Simplification** (10 points)

Which expression is equivalent to $\frac{25(x^2-5)}{(x-\sqrt{5})}$, where $x \neq \sqrt{5}$?

- A. $5(x - 5)$
- B. $5(x + 5)$
- C. $25(x - \sqrt{5})$
- D. $25(x + \sqrt{5})$

Answer:

□

31. **Circle Radius** (10 points)

The equation below defines a circle in the xy -coordinate plane:

$$x^2 + y^2 - 6x - 8y = -16$$

What is the length of the circle's radius?

Answer:

□

32. **Arithmetic Sequence Sum** (10 points)

A sequence of numbers starts at the number a . Each sequential number is k more than the previous number. The sum of the first 13 numbers can be written as $xa + yk$. What is the value of $x + y$?

Answer:

□

33. **Budget Interpretation** (10 points)

Maximum Mail Order Company upgrades its computers each year. The function $P(t) = 36t + 410$ models the company's budget for upgrades, in thousands of dollars, t years after 2004. What is the best interpretation of 36 in this context?

- A. The number of years until the budget is completely spent
- B. The amount budgeted, in thousands of dollars, in the year 2004
- C. The annual increase, in thousands of dollars, in the budget each year after 2004
- D. The number of years in which the budget will increase after 2004

Answer:

□

34. **Algebraic Expression Equivalence** (10 points)

Which expression is equivalent to $\frac{4x^2-3}{2x+\sqrt{3}}$?

- A. $2x - 3$
- B. $2x - \sqrt{3}$
- C. $2x + \sqrt{3}$
- D. $2x + 3$

Answer:

□

35. **y -intercept of Exponential Equation** (10 points)

$$y - 7 = 3^x - 5$$

What is the y -intercept for the given equation?

Answer:

□

36. **Linear Model for Calf Weight** (10 points)

A linear model estimates that the mean weight of a calf in New Zealand, at birth, is 34 kilograms. At 180 days after birth, the mean weight is 272 kilograms. Which function best defines this model, where $f(x)$ is the mean weight of a calf, in kilograms, x days after it is born, for $0 \leq x \leq 180$?

- A. $f(x) = 0.76x + 34$
- B. $f(x) = 0.76x + 180$
- C. $f(x) = 1.32x + 34$
- D. $f(x) = 34x + 272$

Answer:

□

37. **Function Table Value** (10 points)

For the function f , the table shows some values of x and their corresponding values of $f(x)$. If $f(1) = t$,

x	$f(x)$
1	3
3	6
5	9
7	12
9	15

what is the value of $f(3t)$?

Answer:

□

38. **Percent Increase from Chart** (10 points)

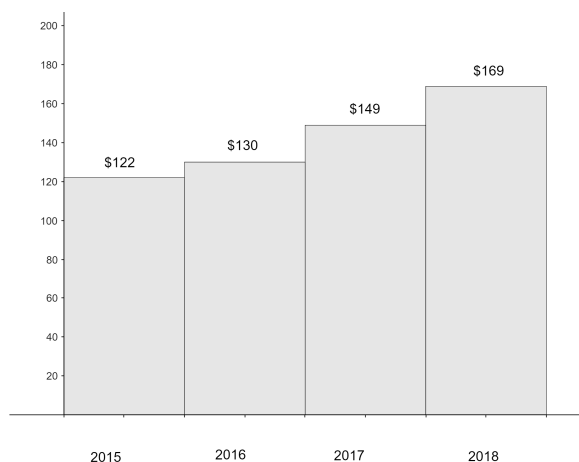


Figure 2: reference attached

The chart shows the average cost of housing, in dollars per square foot, in Nebraska from 2015 to 2018. By approximately what percent did the average cost of housing increase over this period?

- A. 20%
- B. 28%
- C. 39%
- D. 47%

Answer:

□

39. **Bacteria Growth Interpretation** (10 points)

$$p(t) = 2034(3)^{\frac{t}{14}}$$

The given function shows the number of bacteria in a pond after t days since the bacteria began to populate the pond. Which of the following is the best interpretation of $(3)^{\frac{t}{14}}$ in the equation?

- A. The number of bacteria at the beginning of the study
- B. The number of bacteria triples every two weeks
- C. The number of bacteria increases by 3 every two weeks
- D. The number of bacteria in the pond after two weeks

Answer:



40. **Factoring Polynomial** (10 points)

$$12x^4 - 8x^3 - 15x^2$$

The given expression is equal to $x^2(qx + 5)(rx - 3)$, where $q > 0$ and $r > 0$. What is the value of $q + r$?

Answer:



41. **Equation of Parallel Line** (10 points)

Line a is parallel to line b . Line a goes through the points $(-4, 0)$ and $(0, 6)$. If line b goes through the point $(-6, 0)$, what is the equation for line b ?

- A. $y = \frac{2}{3}x - 6$
- B. $y = \frac{2}{3}x + 4$
- C. $y = \frac{3}{2}x + 6$
- D. $y = \frac{3}{2}x + 9$

Answer:



42. **Equilateral Triangle Exterior Angle** (10 points)

The exterior angle of an equilateral triangle is equal to $(8x - 20)^\circ$. What is the value of x ?

- A. 5
- B. 10
- C. 12.5
- D. 17.5

Answer:



43. **Quadratic Equation Solutions** (10 points)

$$(x - 1)(x + 2) = x(x - 3)$$

How many solutions does the given equation have?

- A. 0
- B. 1
- C. 2
- D. 4

Answer:



44. **System of Inequalities Solution** (10 points)

$$y > -2x - 1$$

$$3y < x + 9$$

Which of the following coordinates would be true for the given system of inequalities?

- A. $(-3, 3)$
- B. $(-2, 1)$
- C. $(1, 4)$
- D. $(3, 1)$

Answer:



45. **Infinite Solutions in System of Equations** (10 points)

$$9x - 14y = -3$$

$$ax - by = 6$$

If the given system of equations has infinite solutions, what is the value of b ?

Answer:



46. **Equivalent Forms of Quadratic Function** (10 points)

$$f(x) = 6x^2 - 45x + 75$$

Which of the following equivalent forms of the given function f displays the x -intercepts as constants or coefficients?

A. $f(x) = (2x - 5)(3x - 15)$

B. $f(x) = 3x(2x - 15) + 75$

C. $f(x) = 6\left(x - \frac{15}{4}\right)^2 + \frac{75}{8}$

D. $f(x) = 6(x - 5)\left(x - \frac{5}{2}\right)$

Answer:

□

47. **System with One Real Solution** (10 points)

$$y = 4$$

$$y = x^2 + 12x + n$$

In the given system of equations, n is a positive constant. For what value of n does the system have exactly one real solution?

Answer:

□

48. **Minimum Value in System of Inequalities** (10 points)

$$y \geq 2(x - 3)^2 + 5$$

$$y \leq x + 3$$

If (a, b) is a solution to the given system of inequalities, what is the minimum value of b ?

A. -5

B. -3

C. 3

D. 5

Answer:

□

49. **Bacteria Doubling Problem** (10 points)

Biology students grew a culture of bacteria at a constant temperature of 55°C for 3 days. The students began with 200 spores and the number of spores doubled every 12 hours. Of the equations given, which can be used to correctly calculate the number of spores in the bacteria culture at the end of 3 days?

- A. $y = 200(12)^3$
- B. $y = 200(2)^3$
- C. $y = 200(2)^6$
- D. $y = 200(12)^2$

Answer:



50. **Comparing Means and Standard Deviations** (10 points)

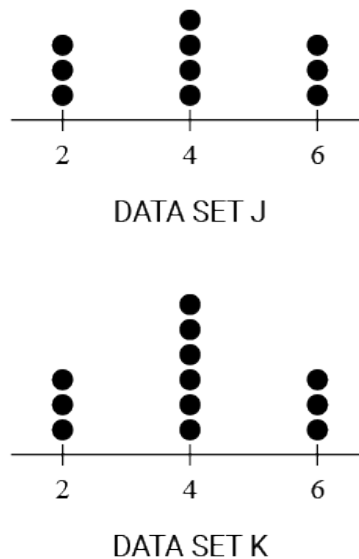


Figure 3: reference attached

Which of the statements best compares the means and standard deviations of Data Set J and Data Set K ?

- A. The mean of Data Set J is less than the mean of Data Set K , and the standard deviations of the two sets are equal.
- B. The mean of Data Set J is greater than the mean of Data Set K , and the standard deviations of the two sets are equal.
- C. The means of the two sets are equal, and the standard deviation of Data Set J is less than the standard deviation of Data Set K .
- D. The means of the two sets are equal, and the standard deviation of Data Set J is greater than the standard deviation of Data Set K .

Answer:



51. **Circumference of a Circle** (10 points)

What is the circumference of the circle with an equation of $x^2 + 6x + y^2 - 4y = 51$?

- A. 8π
- B. 14.28π
- C. 16π
- D. 64π

Answer:



52. **Trigonometric Value in Triangle** (10 points)

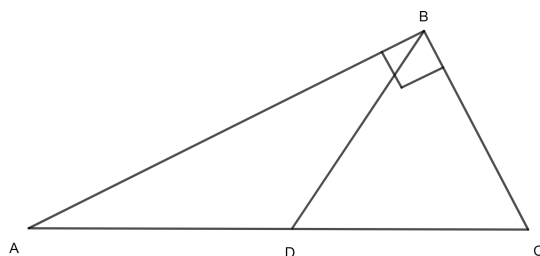


Figure 4: reference attached

In the triangle shown, $\angle ABC = 90^\circ$. Line segment BD is shown with point D lying on line AC . What is the value of $\sin \angle ABD - \cos \angle DBC$?

- A. 0
- B. 30
- C. 45
- D. 90

Answer:



53. **No Solution for System of Equations** (10 points)

$$y = (x - 6)(x + 6)$$

$$y = k$$

In the system of equations shown, k is an integer constant. What is the largest value of k for which the system has no solution?

Answer:



54. **Exponential Function Shift** (10 points)

The base of the exponential function $g(x)$ is k , where k is a positive real number, and k is equivalent to the value of the y -coordinate of the y -intercept of the graph of $g(x)$. Which of the following functions describes the exponential function $g(x - 1)$?

A. $g(x - 1) = k^x$

B. $g(x - 1) = k^{x+1}$

C. $g(x - 1) = k^x + k$

D. $g(x - 1) = k^{x-1} + k$

Answer:

